1. Propositional logic

Proposition: declarative statement that is a declarative statement that is either true or false (but not both)
- if true, then truth values is True (T)
- if false, the truth value is False (F)

Propositional variables (Atoms): used to represent propositions
Ex. P: "Tigers are furry animals"

Compound proposition: formed by combining one or more atoms using logical connectives.

Logical connectives

Negation

P	¬P
T	F
F	T

¬P → "not P"

Conjunction

P	Q	P∧Q
T	T	T
T	F	F
F	T	F
F	F	F

P∧Q → "P and Q"

Disjunction

P	Q	P∨Q
T	T	T
T	F	T
F	T	T
F	F	F

P∨Q → "P or Q"

Exclusive or

P	Q	P⊕Q
T	T	F
T	F	T
F	T	T
F	F	F

P⊕Q → "P or Q but not both"

Conditional statement/implication

P	Q	P→Q
T	T	T
T	F	F
F	T	T
F	F	T

P→Q → "If P then Q"

Bi-conditional statement/(double implication)

P	Q	P↔Q
T	T	T
T	F	F
F	T	F
F	F	T

P↔Q → "P if & only if Q"

Truth assignment of ∀ X: assignin T or F to each atom of X

Tautology: always T **contradiction**: always F

contingency: at least one T & one F

2. Logical Equivalence, Translation between English and propositional logic

Precedence:
1. ¬ 2. ∧ 3. ∨ 4. → 5. ↔ 6. ⊕

Logical equivalence: logically equivalent if P↔Q is a tautology
Notation: P ≡ Q
↳ ≡ means "is logically equivalent to"

contrapositive of P→Q is the implication ¬Q → ¬P
converse of P→Q is the implication Q → P
Counterexamples: examples that prove that a statement is false
Propositional function: statement involving variables, such as "x > 1"

Quantifiers

Existential Quantifier ∃
↳means "there exists"

Universal Quantifier ∀
↳means "for all" or "for each"

∈ → is an element of
∉ → is not an element of

negation of OR-Statement
¬(P∨Q) ≡ ¬P∧¬Q

negation of AND-Statement
¬(P∧Q) ≡ ¬P∨¬Q

negation of an implication
¬(P→Q) ≡ P∧¬Q

3. The laws of logical equivalences & DNF

	Logical equivalence	Name of law
1	P→Q ≡ ¬P∨Q	Implication Law
2	P↔Q ≡ (P→Q)∨(Q→P)	Bi-conditional Laws
3	P↔Q ≡ (P∧Q)∨(¬P∧¬Q)	
4	P∨¬P ≡ T	Negation laws
5	P∧¬P ≡ F	
6	P∨F ≡ P	Identity laws
7	P∧T ≡ P	
8	P∨T ≡ T	Domination laws
9	P∧F ≡ F	
10	P∨P ≡ P	Idempotent laws
11	P∧P ≡ P	
12	¬(¬P) ≡ P	Double negation law
13	P∨Q ≡ Q∨P	commutative laws
14	P∧Q ≡ Q∧P	
15	(P∨Q)∨R ≡ P∨(Q∨R)	Associative laws
16	(P∧Q)∧R ≡ P∧(Q∧R)	
17	P∨(Q∧R) ≡ (P∨Q)∧(P∨R)	Distributive laws
18	P∧(Q∨R) ≡ (P∧Q)∨(P∧R)	
19	¬(P∨Q) ≡ ¬P∧¬Q	De morgan's laws
20	¬(P∧Q) ≡ ¬P∨¬Q	
21	P∨(P∧Q) ≡ P	Absorption laws
22	P∧(P∨Q) ≡ P	
23	P⊕Q ≡ (P∧¬Q)∨(¬P∧Q)	Exclusive-or law

Atom: proposition containing no logical connectives (propositional variable)
Literal: atom or negation of an atom
conjunctive clause: compound proposition that contains only literals and (possibly) the connective ∧, and no atom appears more than once
disjunctive normal form (DNF): a compound proposition is said to be DNF if it is a disjunction of conjuctive clauses

Determining whether a set of compound propositions is consistent

- A set $\{P_1, P_2, \dots, P_n\}$ of compound is called **consistent** if ∃ a truth assignment such that ∀ $P_i \in \{P_1, \dots, P_n\}$, P_i is T. i.e. there exists a truth ass't for which all the propositions are T.
- If $\{P_1, P_2, \dots, P_n\}$ is not consistent, then, for each possible truth assignment, at least one of the propositions P_i is false, in this case, the set $\{P_1, P_2, \dots, P_n\}$ is called **inconsistent**. In this case, the conjunction $P_1 \wedge \dots \wedge P_n$ is a contradiction

Equivalently, a set $\{P_1, P_2, \dots, P_n\}$ is...
- consistent if the conjunction $P_1 \wedge \dots \wedge P_n$ is not a contradiction
- inconsistent if the conjunction $P_1 \wedge \dots \wedge P_n$ is a contradiction

★To test whether the set $\{P_1, P_2, \dots, P_n\}$ is consistent
- grow a tree with root

P_1
P_2
⋮
P_n
← this root corresponds to non-splitting rule for the conjunction $P_1 \wedge \dots \wedge P_n$

- if all paths are dead, then $(P_1 \wedge P_2 \wedge \dots \wedge P_n)$ is a contradiction, which means $(P_1 \wedge P_2 \wedge \dots \wedge P_n)$ can never be true, which means $\{P_1, P_2, \dots, P_n\}$ is inconsistent
- if one path (or more) is alive (complete active path), then these paths give the truth assignments that prove that $(P_1 \wedge P_2 \wedge \dots \wedge P_n)$ can be true, which translates to showing that $\{P_1, P_2, \dots, P_n\}$ is consistent

4. Truth trees

Truth tree: alternative structure for examining all the ways that a compound proposition, say X, can be true.

Branching rules

splitting: a∨b → a | b
non-splitting: a∧b → a, b

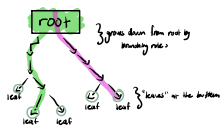
splitting: ¬(a∧b) → ¬a | ¬b (De morgan's)
non-splitting: ¬(a∨b) → ¬a, ¬b

splitting: a→b → ¬a | b (Implication law)
non-splitting: ¬(a→b) → a, ¬b

One more non-splitting rule: ¬¬a → a (Double negation law)

Three more splitting rules:
- a↔b → a, b | ¬a, ¬b (Bi-conditional law: a↔b ≡ (a∧b)∨(¬a∧¬b))
- ¬(a∨b) → ¬a, ¬b | a, ¬b
- a⊕b → a, ¬b | ¬a, b

How to grow a truth tree



- A path from a leaf back to the root is called **inactive**, informally "dead", if it contains an atom as well as that atom's negation, otherwise the path is called **active** or informally "live".
- A path from a leaf back to the root is called **complete** if there are no unchecked propositions on that path.
- Each complete active path tells us one way that will make the root true.
- The tree is **complete** (i.e. "fully grown") when each path from leaf to root is inactive ("dead") or complete

5. Arguments & rules of inference

Argument: set of proposition in which one (called the conclusion) is claimed to follow from the other propositions (called the premises). More specifically, an argument is a compound proposition of the form.

argument: $(P_1 \wedge P_2 \wedge \dots \wedge P_k) \to Q$
($P_1, \dots, P_k$ are the premises; Q is the conclusion)

so $(P_1 \wedge P_2 \wedge \dots \wedge P_k) \to Q$ is a valid argument if $(P_1 \wedge P_2 \wedge \dots \wedge P_k) \to Q$ is a tautology

Rule of inference	tautology	Name
P, P→Q ∴ Q	(P∧(P→Q))→Q	modus ponens
¬Q, P→Q ∴ ¬P	(¬Q∧(P→Q))→¬P	modus tollens
P→Q, Q→R ∴ P→R	((P→Q)∧(Q→R))→(P→R)	hypothetical syllogism
P∨Q, ¬P→Q ∴ Q	((P∨Q)∧(¬P))→Q	disjunctive syllogism
P ∴ P∨Q	P→P∨Q	addition
P∧Q ∴ P	(P∧Q)→P	simplification
P, Q ∴ P∧Q	(P∧(Q))→(P∧Q)	conjunction
P∨Q, ¬P∨R ∴ Q∨R	((P∨Q)∧(¬P∨R))→(Q∨R)	resolution

Rule of inference	Name
∀x, P(c) ∴ P(c)	Universal instantiation
P(c) is true for arbitrary c ∴ ∀x, P(x)	Universal generalization
∃x such that P(x) ∴ P(c) for some c	Existential instantiation
P(c) for some c ∴ ∃x such that P(x)	Existential generalization

P(x): x > 3
- P(x) is not a proposition
- variable x is the subject of the statement
- P is the predicate
- the values of the variable x come from a domain of discourse or simply from the domain.
- We can think of P(x) as the value of a propositional function P at x, where x is assigned a particular value from the domain of discourse.
- Once we assign a particular value to x, we obtain a proposition.

Universal Quantification of P(x) is the statement: ∀x, P(x).
Existential Quantification of P(x) is the statement: ∃x, such that P(x).

6. Knights & Knaves Puzzles

- If an inhabitant of the Island of Logic is a statement, whether it is a true statement or a lie depends on the truth value of the entire statement as a whole.
- Any statement made by a Knight is T.
- Any statement made by a Knave is F.

Truth table method
1) For each inhabitant X, define an "I am a Knight" atom for X
2) Translate all speakers' statements into compound propositions
3) Construct a truth table that includes all speakers' "I am a Knight" atoms and all speakers' statements
4) The rows of the truth table correspond to an exhaustive list of possibilities
5) In each row, check whether all speakers' statements' truth values are compatible with their type - that row is T, then the truth assignment of that row is a possible scenario on the Island of Knights and Knaves.
6) Some puzzles might have more than one possible solution (more than one row that makes sense for the Island).
7) Sometimes, the conclusions we can make only answer one aspect of the puzzle (maybe we don't know if A is a Knight or a Knave, but we still know B is a Knave, etc.)

7. Proof Strategies

Theorem: mathematical statement that is true

Direct proof

To prove a conditional statement, such as P→Q, using a direct proof, we will
- Assume P is true
- Then, step-by-step, show that Q must follow from P.

Basically assume P is true so that Q is true to avoid (T→F) (False)

Notation: The integers, denoted $\mathbb{Z}$, is the set $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2 \dots\}$
Definition:
- An integer $n \in \mathbb{Z}$ is called **even** if $\exists k \in \mathbb{Z}$ such that $n = 2k$.
- An integer $n \in \mathbb{Z}$ is called **odd** if $\exists k \in \mathbb{Z}$ such that $n = 2k+1$

Indirect proof

To prove a conditional statement, such as P→Q, using an indirect proof, also called a proof by contraposition, we will
- Assume ¬Q is true (equivalently, assume Q is false).
- Then, step-by-step, show that ¬P must follow from ¬Q

P→Q, ¬Q→¬P

Divisibility

Definition: Let $n \in \mathbb{Z}$ and let d be nonzero integer. We say d divides n, or that n is divisible by, if $\exists q \in \mathbb{Z}$ such that $n = qd$.

Notation: If d divides n, then we write d|n. If d does not divide n, then we write d∤n.

Proof by cases

To prove a statement, such as P→Q, using a proof by cases, we will
- Break P up into cases $P_1, P_2, \dots, P_n$ such that $P \equiv P_1 \vee P_2 \vee \dots \vee P_n$.
- Then, prove each of the conditional statements

$P_1 \to Q$
$P_2 \to Q$
⋮
$P_n \to Q$

using whichever strategy seems best

- Because the cases $P_1, P_2, \dots, P_n$ capture all possible ways to make P true, proving all these cases using conditional statements is equivalent to proving P→Q
- Because $P \equiv P_1 \vee P_2 \vee \dots \vee P_n$, it follows that $(P_1 \to Q) \wedge (P_2 \to Q) \wedge \dots \wedge (P_n \to Q) \equiv (P_1 \vee \dots \vee P_n) \equiv P \to Q$

8. Proof Strategies

Proof of equivalence

To prove a statement, such as P↔Q, using a proof of equivalence, we will
- prove the conditional statement P→Q, and
- prove the converse Q→P

A proof of equivalence simply takes care of each "direction" of the bi-conditional statement individually

Proof by contradiction

To prove a statement, such as P, using a proof by contradiction, we will
- Assume ¬P is true (equivalently, assume P is false)
- Then, step-by-step, show contradiction always follows from ¬P. i.e. prove ¬P→F

A proof by contradiction is akin to the truth tree method of putting ¬P at the root and discovering that all paths are "dead" (meaning lead to contradictions), this tells us that ¬P can never be true, which is a way to know that P must always be true.

Proof by contradiction Vs. Indirect (By contraposition)

To prove a conditional statement, such as P→Q, using a proof by contradiction, we will
- Assume the entire conditional statement is false, that is, assume ¬(P→Q) is true. Equivalently, we assume P∧¬Q is true.
- Then, step-by-step, we show that contradiction always follows when we assume both P and ¬Q are true.

In contrast, if we prove P→Q using an indirect proof (by contraposition), we will
- Assume ¬Q is true.
- Then, prove that ¬P must follow from ¬Q.

Proof by contradiction of the existence of an irrational number

Definition: A real number x is called **Rational** if $\exists n, d \in \mathbb{Z}$ with $d \neq 0$ s.t. $x = \frac{n}{d}$.

A real number x that is not rational is called **Irrational**.
$\forall n, d \in \mathbb{Z}$ with $d \neq 0$, $x \neq \frac{n}{d}$.

Rigorous proofs involving sets

Let A & B be subsets of a universal set U.
To give a Rigorous proof that A ⊆ B we must prove:
$\forall x (x \in A) \to (x \in B)$ is true.
we can prove this directly as follows:
- Let $x \in U$ be arbitrary element of the universal set.
  Assume $x \in A$.
- Then, step-by-step, prove that $x \in B$ must be true.

We can also prove the implication $(x \in A) \to (x \in B)$ using an indirect proof, a proof by contradiction, a proof by cases, or some suitable combination of Proof Strategies.

9. Basics of set theory

Set: a well-defined unordered collection of objects called elements

* A and B are equal ($A=B$) if $\forall x, (x\in A) \leftrightarrow (x\in B)$

$\mathbb{N} \to \{0,1,2,3,\dots\}$ → natural numbers
$\mathbb{Q} \to \{\frac{m}{n} : m,n\in\mathbb{Z}, n\neq 0\}$ → rational numbers
$\mathbb{R} \to \{\text{all real numbers}\}$ → real numbers

$\mathbb{Z} \to \{\dots,-2,-1,0,1,2,3,\dots\}$ → integers
$\mathbb{Z}^+ \to \{n\in\mathbb{Z} : n>0\}$ → positive integers
$\mathbb{Z}^- \to \{n\in\mathbb{Z} : n<0\}$ → negative integers

Empty set: set with no elements, denoted $\emptyset$

Universal set: denoted $\mathcal{U}$, is the set of all objects under consideration.

Subset: denoted $(A\subseteq B)$ if $\forall x, (x\in A)\to(x\in B)$ is true.

Proper subset: written $(A\subset B)$ if $A\subseteq B$ and $A\neq B$

* If a set A has exactly n distinct elements (for $n\in\mathbb{N}$), then A is called finite and the cardinality of A is n (it's "size"). The cardinality of a set A is denoted $|A|$.

* Let A be a set. The power set of A is the set of all subsets of A, denoted $\mathcal{P}(A)$. ie $\mathcal{P}(A) = \{S : S\subseteq A\}$.

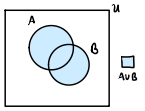
* Let A and B be sets. The Cartesian product "A cross B", denoted $A\times B$, is the set of all ordered pairs (a,b) such that $a\in A$ and $b\in B$.
ie $A\times B = \{(a,b) : a\in A, b\in B\}$
cardinality: $|A\times B| = |A||B|$

* Cartesian product is not associative.

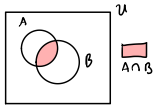
10. Set operations & set identities

Venn diagrams: graphical method for depicting sets & set operations.

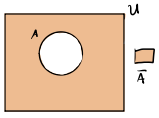
Union
of sets A & B, denoted $A\cup B$
$A\cup B = \{x\in\mathcal{U} : (x\in A)\vee(x\in B)\}$



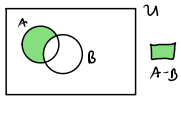
Intersection
of sets A and B, denoted $A\cap B$, is the set
$A\cap B = \{x\in\mathcal{U} : (x\in A)\wedge(x\in B)\}$



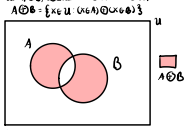
complement
of a set A (with respect to an understood universal set $\mathcal{U}$)
$\bar{A} = \{x\in\mathcal{U} : x\notin A\}$



difference
("A minus B"), denoted $A-B$
$A-B = \{x\in\mathcal{U} : (x\in A)\wedge(x\notin B)\}$



symmetric difference
sum of A & B, denoted $A\oplus B$, is the set
$A\oplus B = \{x\in\mathcal{U} : (x\in A)\oplus(x\in B)\}$



* $A\oplus B = A\cup B - A\cap B$

* Sets A and B are disjoint if $A\cap B = \emptyset$

Set identity: an equation involving sets and set operations that holds true, no matter which particular sets we consider.

Membership table: similar to truth tables, 2^n rows, n locations.

Table of important set identities

	Set identity	MAT 1348 Name of law
1	$A\cup\emptyset = A$	Identity laws
2	$A\cap\mathcal{U} = A$	
3	$A\cup\mathcal{U} = \mathcal{U}$	Domination laws
4	$A\cap\emptyset = \emptyset$	
5	$A\cup A = A$	Idempotent laws
6	$A\cap A = A$	
7	$\overline{(\bar{A})} = A$	Double complementation law
8	$A\cup B = B\cup A$	Commutative laws
9	$A\cap B = B\cap A$	
10	$(A\cup B)\cup C = A\cup(B\cup C)$	Associative laws
11	$(A\cap B)\cap C = A\cap(B\cap C)$	
12	$A\cap(B\cup C) = (A\cap B)\cup(A\cap C)$	Distributive laws
13	$A\cup(B\cap C) = (A\cup B)\cap(A\cup C)$	
14	$\overline{A\cup B} = \bar{A}\cap\bar{B}$	De morgan's laws
15	$\overline{A\cap B} = \bar{A}\cup\bar{B}$	
16	$A\cup(A\cap B) = A$	Absorption laws
17	$A\cap(A\cup B) = A$	
18	$A\cup\bar{A} = \mathcal{U}$	Complement laws
19	$A\cap\bar{A} = \emptyset$	
20	$A - B = A\cap\bar{B}$	Difference law
21	$A\oplus B = (A-B)\cup(B-A)$	Symmetric difference laws
22	$A\oplus B = (A\cup B) - (A\cap B)$	

11. Functions

Function consists of:
- Set A called domain
- Set B called codomain, &
- "rule" f that assigns to each element $a\in A$ exactly one element $b\in B$.

We write $f(a)=b$ if b is the unique element of B assigned by the function f to the element $a\in A$.

- For each $a\in A$, the element $f(a)$ is called the image of f at a or the value of f at a. In particular, $f(a)\in B$.
- For each subset $S\subseteq A$, we may consider the set $f(S) := \{f(a) : a\in S\}$
  The set $f(S)$ is called the image of the set S under f
- In particular, the image of the domain A under f is also called the range of f, namely the set $f(A) = \{f(a) : a\in A\}$.
- Note: $\forall S\subseteq A$, we have $f(S)\subseteq B$.
- For each $b\in B$, we may consider the set $f^{-1}(b) := \{a\in A : f(a)=b\}$
  The set $f^{-1}(b)$ is called the preimage of b under f.
- Note: $\forall b\in B$, we have $f^{-1}(b)\subseteq A$.

Injective or one-to-one

$\forall a_1, a_2\in A$, the implication $(f(a_1)=f(a_2)) \to (a_1=a_2)$ is true
contrapositive: $\forall a_1, a_2\in A, (a_1\neq a_2)\to(f(a_1)\neq f(a_2))$

proof
- Let a_1, a_2 assume $f(a_1)=f(a_2)$
- prove $a_1=a_2$ must be true

If $\exists f: A\to B$, then $|A|\leq|B|$ (f is injective)
If $|A|>|B| \to \forall f: A\to B$, f is not injective

Surjective or onto

$\forall b\in B, \exists a\in A$ such that $f(a)=b$

$\forall b\in B, f^{-1}(b)\neq\emptyset$.

for each element of the codomain B is the image of at least one element of A.

proof
- Let $b\in B$ be an arbitrary element of the codomain of f.
- Construct (figure out / reverse-engineer) an element $a\in A$, in terms of b.
  (difficult if fun is not injective)

If $\exists f: A\to B$, then $|A|\geq|B|$ (f is surjective)
If $|B|>|A|$, then there does not exist a surjective function from A to B

12. Functions: Bijections, compositions, & inverse functions

floor function

$\lfloor\ \rfloor : \mathbb{R}\to\mathbb{Z}$ is defined by
$\lfloor x\rfloor = \max\{n\in\mathbb{Z} : n\leq x\}$
Ex: $\lfloor 1.8\rfloor = 1$

Bijection

$f: A\to B$ is a bijection if
f is injective & surjective

Identity function on A

$id_A : A\to A$
$\forall x\in A, id_A(x) = x$
* is a bijection

composition

$f: A\to B$ $g: C\to D$
if $f(A)\subseteq C$ then $g\circ f: A\to D$
defined by $(g\circ f)(a) = g(f(a))$ for all $a\in A$
* $\forall a\in A$, we need $f(a)$ to be an element of g's domain; otherwise, $g(f(a))$ would not be defined

Inverse

$f: A\to B$

$f^{-1}\circ f = id_A$ and $f\circ f^{-1} = id_B$

* for all $a\in A$, $b\in B$, $f^{-1}(b)=a \leftrightarrow f(a)=b$

- Not every function has an inverse.
- If a fun has an inverse then it's invertible
- If f is invertible, then its inverse is unique meaning there is one and only one function from B to A whose compositions with f give the respective identity functions.
- Let $f: A\to B$ be a function
  f is invertible $\leftrightarrow$ f is a bijection.

* infinite set S countable if $|S| = |\mathbb{N}|$
* There is no bijection from $\mathbb{R}$ to $\mathbb{N}$. Therefore, the set of real numbers is called uncountable.

13. Relations

A (Binary) relation from A to B is a subset of $A\times B$.
A (Binary) relation on a set A is a relation from A to itself. ie a subset of $A\times A$.

Notations
- $(1,x)\in R$, means "1 is related to x by R,"
- $(2,x)\notin R$, means "2 is not related to x by R."

Reflexive
a relation R on set A

reflexive: $\forall a\in A, (a,a)\in R$
↳ for every element a in the set, the ordered pair (a,a) must be included in the relation.

14. Equivalence Relations, Classes & partitions

Equivalence relation on a set A is a relation that is reflexive, symmetric, and transitive.

* Logical equivalence $\equiv$ is a relation on A given by the rule: $\forall P,Q\in A$, $P\equiv Q \leftrightarrow (p\leftrightarrow q\equiv T)$

Given an equivalence relation R on A, for each element $a\in A$, we define the Equivalence class of a with respect to R as follows: $[a]_R = \{x\in A : xRa\}$
= set of all elements of A which are related to a by R

Congruence modulo n ($\equiv (\mathrm{mod}\ n)$) Let $n\in\mathbb{Z}^+$ be a fixed positive integer, called the modulus.
We define a relation on $\mathbb{Z}$, called congruence modulo n, denoted $\equiv(\mathrm{mod}\ n)$, by the following rule: $\forall x,y\in\mathbb{Z}$, $x\equiv y\ (\mathrm{mod}\ n) \leftrightarrow n$ divides $x-y$

$\forall x,\ (x \in A) \to ((x,x) \in R)$

or

$\forall x,\ (x \in A) \to (x R x)$

Symmetric

$\forall x,y \in A,\ ((x,y) \in R) \to ((y,x) \in R)$

or

$\forall x,y \in ,\ (x R y) \to (y R x)$

Symmetric: $(a,b) \in R \to (b,a) \in R$

If a is related to b, then b must also be related to a.

Antisymmetric: $(a,b) \in R$ and $(b,a) \in R \to a=b$

If both (a,b) and (b,a) are in the relation, then a & b must actually be the same element

Transitive: $(a,b) \in R$ and $(b,c) \in R \to (a,c) \in R$

If a is related to b, and b is related to c, then a must also be related to c

Antisymmetric

$\forall x,y \in A,\ ((x,y) \in R \wedge (y,x) \in R) \to (x=y)$

or

$\forall x,y \in A,\ (x R y \wedge y R x) \to (x=y)$

or

$\forall x,y \in A,\ (x \neq y) \to ((x,y) \notin R \text{ or } (y,x) \notin R)$

Transitive

$\forall x,y,z \in A,\ ((x,y) \in R \wedge (y,z) \in R) \to ((x,z) \in R)$

or

$\forall x,y,z \in A,\ (x R y \wedge y R z) \to (x R z)$

* A relation R on a set A is called an Equivalence Relation if R is reflexive, symmetric, and transitive

17. The pigeonhole principle

Theorem

$k \in \mathbb{Z}^+$

If $k+1$ objects are placed into k boxes, then there is at least one box that contains at least 2 objects

Pigeonhole principle

If a flock of $k+1$ pigeons flies into a set of k pigeonholes, then at least one pigeonhole contains at least two pigeons.

The ceiling function (function from $\mathbb{R}$ to $\mathbb{Z}$)

for $x \in \mathbb{R}$, the ceiling of x, denoted $\lceil x \rceil$, is defined as follows:

$$\lceil x \rceil = \min \{n \in \mathbb{Z} : x \leq n\} \quad \text{"rounding up"}$$

Theorem

(The generalized pigeonhole principle) Let k and N be positive integers.

If N objects are placed into k boxes, then there is at least one box that contains at least $\lceil \frac{N}{k} \rceil$ objects.

Theorem (The generalized pigeonhole principle)

we will prove GPP by contradiction. GPP says: (we place N objects into k boxes) $\to$ (at least one box contains at least $\lceil N/k \rceil$ objects)

$P \to Q$

Assume the negation of GPP is true. That is, assume $\neg(P \to Q) \equiv P \wedge \neg Q$

Assume we place N objects into k boxes and it is not the case that at least one box contains at least $\lceil N/k \rceil$ objects. Then each box contains at most $\lceil N/k \rceil - 1$ objects.

$$\Rightarrow (\text{total \# of objects}) = (\text{\# objects in box 1}) + (\text{\# objects in box 2}) + \dots + (\text{\# objects in box } k)$$

$$\leq \lceil N/k \rceil - 1 + \lceil N/2 \rceil - 1 + \dots + \lceil N/k \rceil - 1$$

$$= k(\lceil N/k \rceil - 1)$$

$$< k(N/k + 1 - 1)$$

$$= N \quad \lightning \text{ this is a contradiction since total \# objects} = N \text{ and } i < N.$$

$\therefore$ the negation of GPP must be false since it leads to contradiction

$\therefore$ the GPP must be true

If $x \equiv y \pmod n$, we say "x is congruent to y modulo n".

* Each integer m is congruent modulo n to its remainder upon division by n.

Theorem: Let n be a positive integer (the modulus). Then the relation $\equiv \pmod n$ of congruence modulo n is an equivalence relation on $\mathbb{Z}$.

we need to prove $\equiv \pmod n$ is reflexive, symmetric, and transitive

Partition of a set A is a collection $P = \{S_1, S_2, \dots\}$ of subsets $S_i \subseteq A$ such that the following three properties hold:

i. $S_i \neq \emptyset$ for all i (S_i are non-empty subsets of A)

ii. $A = S_1 \cup S_2 \cup \dots$ (union of all S_i is all of A)

iii. $S_i \cap S_j = \emptyset$ for all $i \neq j$ (pairwise disjoint)

check examples lec 16 à reviser for sure

15. Counting: the product rule

suppose a procedure can be broken down into a sequence of k tasks $T_1, T_2, \dots, T_k$ such that

- there are n_1 ways to carry out task T_1
- and, for $i = 2, \dots, k$, there are n_i ways to carry out task T_i after each of the tasks $T_1, T_2, \dots, T_{i-1}$ has been done in a certain way.

There are $(n_1)(n_2)\cdots(n_k)$ ways to carry out the procedure.

16. Counting: permutations, combinations, PIE & The Sum Rule

Permutation: ordered arrangement of some (or all) elements of a finite set.

Definition. Let n, k be integers such that $n \geq k \geq 0$.

A k-permutation of a set S of cardinality n is an ordered arrangement of k distinct elements of S.

Notation:

$P(n,k)$ = # of k-permutations of an n-element set

$P(n,n)$ = # of n-permutations of an n-element set

Factorials

for each $n \in \mathbb{N}$ "n factorial" is denoted $n!$ and defined recursively as follows: For $n = 0$: $0! = 1$, For $n \geq 1$: $n! = n(n-1)! = n(n-1)(n-2)\cdots(2)(1)$

Combinations

$n \geq k \geq 0$ be integers.

A k-combination of A set S of cardinality n is a k-element subset of S.

Notation. $C(n,k)$ = # of k-combinations from an n-element set = # of subsets T of S such that $|T| = k$ and $|S| = n$

Theorem

Let $n, k \in \mathbb{N}$ with $n \geq k \geq 0$

Then $P(n,k) = C(n,k) \cdot P(k,k)$ consequently, $C(n,k) = \frac{n!}{k!(n-k)!}$

Theorem

Let A and B be sets, Then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

suppose a procedure can be broken down into k tasks $T_1, T_2, \dots, T_k$ such that

- for $i = 1, 2, \dots, k$, the entire procedure can be done by task T_i which can be carried out in one of n_i ways
- and the tasks $T_1, T_2, \dots, T_k$ are mutually exclusive i.e. none of the tasks share a common way to accomplish the procedure

Then there are $n_1 + n_2 + \dots + n_k$ ways to carry out the procedure

Let k, r be positive integers.

Suppose some objects are to be placed into k boxes.

Question

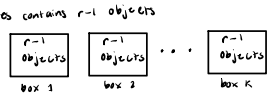
what is the minimum number N of objects that we need to place into k boxes in order to guarantee that at least one box contains at least r objects?

Answer

The minimum number N that is needed to guarantee that at least one box contains at least r objects is $N = k(r-1) + 1$

Explanation

With only $k(r-1)$ objects, we could end up in the worst-case scenario: each of the k boxes contains $r-1$ objects



$\therefore k(r-1)$ objects is too few

By G.P.P., at least one box is guaranteed to contain at least $\lceil N/k \rceil$ objects.

with $N = k(r-1) + 1$, we get

$$\lceil N/k \rceil = \left\lceil \frac{k(r-1)+1}{k} \right\rceil = \lceil r - 1 + 1/k \rceil$$

$$= r \text{ objects}$$

$\therefore N = k(r-1)+1$ is enough

$\therefore N = k(r-1)+1$ is exactly the minimum threshold to guarantee that at least one box contains at least r objects.

Ex 17.7

$N = 12(3-1)+1$

$= 12 \cdot 2 + 1$

$= 24 + 1$

$= 25$

Ex 17.8

26^2

$N = 26^2 \cdot (2-1) + 1$

$= 26^2 + 1$

18. Proof by induction

weak induction

The principle of mathematical induction (PMI) – weak induction

Setup: suppose, for each $n \in \mathbb{Z}$, $P(n)$ is a statement that depends on n.

Let $b \in \mathbb{Z}$ be a given Base value.

Goal: prove $P(n)$ is true for all integers $n \geq b$.

Strategy + basis of induction (B.I.): Prove $P(b)$ (the "base case")

** Induction step (I.S.): Show $P(k) \Rightarrow P(k+1)$ holds true for an arbitrary integer $k \geq b$.

*** Induction hypothesis (I.H.):
Let k be an arbitrary integer such that $k \geq b$
Assume $P(k)$ is true (goal: prove $P(k+1)$)

Reasoning:
- we prove $P(b)$ holds
- we prove $\forall\, k \geq b$, $P(k) \Rightarrow P(k+1)$.

$\therefore P(b)$, $P(b) \Rightarrow P(b+1)$, $P(b+1) \Rightarrow P(b+2)$, $P(b+2) \Rightarrow P(b+3)$, $\ldots$

$\therefore P(b), P(b+1), P(b+2), P(b+3), \ldots$

By PMI, this proves $P(n)$ holds for all integers $n \geq b$

Ex 18.1. Use weak induction to prove that the following formula holds for all integers $n \geq 1$:

$$\underbrace{1 + 2 + \ldots + n}_{\sum_{i=1}^{n} i} = \frac{n(n+1)}{2}$$

for each integer $n \geq 1$, define the statement

$P(n)$: $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ Note: $P(n)$ must be a statement

this notation denotes a completely declarative sentence

$P(n)$: "the sum of the first n positive integers equals $\frac{n(n+1)}{2}$"

* Base case ($b=1$): $\sum_{i=1}^{1} i = 1$ and $\frac{(1)(1+1)}{2} = 1$ $\therefore P(1)$ is true

** Induction step:
we will prove $P(k) \Rightarrow P(k+1)$ holds for an arbitrary $k \geq 1$.

*** Induction Hypothesis:
Let $k \in \mathbb{Z}$ such that $k \geq 1$.
Assume $P(k)$ is true.
i.e. assume $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ (pf: prove $P(k+1)$ follows from this / i.e. prove $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$)

Then $\sum_{i=1}^{k+1} i = \underbrace{1+2+\ldots+k}_{\sum_{i=1}^{k} i} + (k+1)$

$= \frac{k(k+1)}{2} + (k+1)$ (by I.H.)

$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$

$= \frac{(k+1)(k+2)}{2}$

$\therefore \sum_{i=1}^{k+1} i = \frac{(k+1)(k+1+1)}{2}$

so $P(k+1)$ holds! Induction step complete!

Summary:
we proved $P(1)$
we proved $P(k) \Rightarrow P(k+1)$ for an arbitrary integer $k \geq 1$
By the PMI, it follows that $P(n)$ is true for all integers $n \geq 1$.
$\therefore \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ holds $\forall\, n \geq 1$.

Quest: Evaluate $\sum_{i=1}^{n} i^2$ for several small values of n

try to find a pattern (the answer to the sum as a function of n). Prove your answer by induction

Ex 18.2 Let $a_0, a_1, a_2, a_3, \ldots$ be a sequence of numbers defined according to the following relation:

$a_0 = 1$
for each integer $n \geq 1$ $a_n = 5(a_{n-1})^2$ ← this is the definition of the sequence; recursive rule applies only for $n \geq 1$

using the recurrence relation, compute the values for a_1, a_2, a_3, and a_4:

$a_0 = 1$ (by def.)
$a_1 = 5(a_0)^2 = 5(1)^2 = 5$
$a_2 = 5(a_1)^2 = 5(5)^2 = 5^3$
$a_3 = 5(a_2)^2 = 5(5^3)^2 = 5^7$
$a_4 = 5(a_3)^2 = 5(5^7)^2 = 5^{15}$

In general, it looks like $\boxed{a_n = 5^{2^n - 1}}$ Let's prove this!

holds for all integers $n \geq 0$

Proof

For each integer $n \geq 0$, let $P(n)$ denote the following proposition:
$P(n)$: "$a_n = 5^{2^n - 1}$"

* B.I. $b=0$ $P(0)$ says "$a_0 = 5^{2^0 - 1}$"
According to the recurrence relation's definition $a_0 = 1$ ✓ equal!
For $n=0$, our formula gives $5^{2^0 - 1} = 5^{1-1} = 5^0 = 1$
$\therefore P(0)$ is true

** IS. We will prove $P(k) \Rightarrow P(k+1)$ holds for an arbitrary integer $k \geq 0$.

*** IH. Let $k \in \mathbb{Z}$ such that $k \geq 0$.
Assume $P(k)$ is true.
i.e. assume $a_k = 5^{2^k - 1}$ (goal: prove $P(k+1)$ follows from IH / i.e. prove $a_{k+1} = 5^{2^{k+1} - 1}$)

Then $a_{k+1} = 5(a_k)^2$ (by the recurrence relation)
$= 5[5^{2^k - 1}]^2$ (by the I.H. !)
$= 5 \cdot 5^{2 \cdot 2^k - 2}$ "
$= 5 \cdot 5^{2^{k+1} - 2}$ "
$= 5^1 \cdot 5^{2^{k+1} - 2}$ "
$= 5^{2^{k+1} - 2 + 1}$ "
$= 5^{2^{k+1} - 1}$ $\therefore a_{k+1} = 5^{2^{k+1} - 1}$ so $P(k+1)$ holds!

Induction step complete!

Summary:
we proved $P(0)$
we proved $P(k) \Rightarrow P(k+1)$, $\forall\, k \geq 0$.
$\therefore P(n)$ holds true $\forall\, n \geq 0$

Strong induction

setup: suppose, for each $n \in \mathbb{Z}$, $P(n)$ is a statement that depends on n.

Let $b \in \mathbb{Z}$ be a given Base value

goal: prove $P(n)$ is true for all integers $n \geq b$.

strategy + basis of induction (B.I.): prove $P(b)$ and $P(b+1)$ and ... and $P(b+j)$ all hold, where j is some integer s.t. $j \geq 0$.

** Strong induction step (I.S.)
Prove $[P(b) \wedge P(b+1) \wedge \cdots \wedge P(k)] \Rightarrow P(k+1)$
for an arbitrary int. $k \geq b+j$.

*** strong induction hypothesis (I.H.)
Let k be an arbitrary integer such that $k \geq b+j$.
Assume $P(b)$ and ... and $P(k)$ all hold true. (goal: prove $P(k+1)$).

Reasoning:
- we prove $P(b), \ldots, P(b+j)$
- we prove $[P(b) \wedge \ldots \wedge P(k)] \Rightarrow P(k+1)$ for arbitrary integer $k \geq b+j$.

$P(b), P(b+1) \Rightarrow P(b+2)$, $P(b), P(b+1), P(b+2) \Rightarrow P(b+3)$, $\ldots$

$\therefore P(b), P(b+1), P(b+2), P(b+3), \ldots$

By PMI, this proves $P(n)$ holds for all integers $n \geq b$.

Ex 18.3. Consider the infinite integer sequence $\{x_n\}_{n=1}^{\infty}$ defined recursively as follows:

$x_1 = 2$
$x_2 = -12$
for all integers $n \geq 3$, $x_n = 2x_{n-1} + 3x_{n-2}$ ← definition of this sequence; this recurrence rule applies only for $n \geq 3$

using strong induction, prove that x_n is divisible by 6 for all integers $n \geq 1$.

$x_1 = 2$ (by def.) ✓
$x_2 = -12$ (by def.) ✓
$x_3 = 2x_2 + 3x_1$ (by def.) $= 2(-12) + 3(2) = -18$ ✓

for each integer $n \geq 1$, define the statement $P(n)$: $6 \mid x_n$.
goal: prove $P(n)$ holds for all integers $n \geq 1$
i.e. prove $6 \mid x_n, \forall\, n \geq 1$

* B.I. $(b=1)$ $x_1 = 2$ by def. $= 6(1) \therefore 6 \mid x_1$ so $P(1)$ holds
$(b=2)$ by rec $x_2 = 2x_1 + 3x_0 = 2(-12) + 3(2) = -18$
$= 6(-3)$
$\therefore 6 \mid x_2$ so $P(2)$ holds

** IS. We will prove $P(1) \wedge \ldots \wedge P(k) \Rightarrow P(k+1)$ for arbitrary $k \geq 1$.

*** IH. Let $k \in \mathbb{Z}$ such that $k \geq 1$
Assume $P(1)$ and ... and $P(k)$ all hold
i.e. assume $6 \mid x_1$, $6 \mid x_2$, ..., and $6 \mid x_k$
(goal: prove $P(k+1)$ follows from this / i.e. prove $6 \mid x_{k+1}$)
then $\exists\, m_1, \ldots, m_k \in \mathbb{Z}$ s.t. $x_1 = 6m_1, \ldots, x_k = 6m_k$
(by def. of divides)

Since $k+1 \geq 2$, the recurrence rule holds for x_{k+1}:
$\Rightarrow x_{k+1} = 2x_k + 3x_{k-1}$ Note: if $k=2$, $6 \mid x_{k-1}$ (by I.H.)
$= 2(6m_k) + 3(6m_{k-1})$ ($\in \mathbb{Z}$) $= 6(2m_k + 3m_{k-1})$ ($\in \mathbb{Z}$)
if $k \geq 2$, $k+1 \geq 3$ so the rule holds with range for strong I.H.

$\Rightarrow 6 \mid x_{k+1}$ which completes the induction step!

Principle of mathematical induction

	Proof by weak induction	Proof by strong induction
To prove:	$\forall\, a \geq b$, such that $n \geq b$, $P(n)$ holds	$\forall\, a \geq b$, such that $n \geq b$, for some integer $j \geq 0$, $P(n), P(n+1), \ldots, P(n+j)$ all hold
Base case	prove $P(b)$.	prove each of $P(b), P(b+1), \ldots, P(b+j)$
Ind. Step	prove $P(k) \Rightarrow P(k+1)$ holds for an arbitrary integer $k \geq b$	prove $[P(b) \wedge P(b+1) \wedge \cdots \wedge P(k)] \Rightarrow P(k+1)$ holds for an arbitrary integer $k \geq b+j$
Ind. Hyp.	Let $k \geq a$, such that $k \geq b$. Assume $P(k)$ holds true.	Let $k \geq b+j$, such that $k \geq b$. Assume $P(b), P(b+1)$ and ... and $P(k)$ all hold true.
goal of IS	Prove $P(k+1)$ follows from IH	Prove $P(k+1)$ follows from IH

19. Binomial coefficients & the binomial theorem

Recall: $C(n,k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$

$C(n,k)$: # k-combos of an n-element set.

called a "binomial coefficient" reads as "n choose k"

Simple observations on binomial coefficients

$\binom{n}{0} = \frac{n!}{0!(n-0)!} = \frac{n!}{0!\, n!} = \frac{1}{0!} = \frac{1}{1} = 1$ $\quad \binom{n}{1} = \frac{n!}{1!(n-1)!} = \frac{n \cdot (n-1)!}{(1)(n-1)!} = \frac{n}{1} = n$

For any $k \in \{0, 1, \ldots, n\}$,

$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-(n-k))!(n-k)!} = \frac{n!}{(n-k)!(n-(n-k))!} = \binom{n}{n-k}$

Ex $\binom{5}{3} = \binom{5}{5-3} = \binom{5}{2}$

Also,
$\binom{n}{n} = \binom{n}{n-n} = \binom{n}{0} = 1$ $\quad \binom{n}{n-1} = \binom{n}{n-(n-1)} = \binom{n}{1} = n$

Pascal's identity & Pascal's triangle

Theorem (Pascal's identity) Let n and k be integers such that $n \geq k+1$ and $k \geq 0$.

Then $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$

Proof of Pascal's Identity

Let $n, k \in \mathbb{Z}$ s.t. $n \geq k+1$ and $k \geq 0$.

LS $= \binom{n}{k} + \binom{n}{k+1} = \frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-(k+1))!}$

$= \frac{n!(k+1)}{k!(n-k)(n-k-1)!(k+1)} + \frac{n!(n-k)}{(k+1)k!(n-k-1)!(n-k)}$

$= \frac{n!(k+1) + n!(n-k)}{(k+1)!(n-k)!}$

$= \frac{n![k+1+n-k]}{(k+1)!(n-k)!} = \frac{n!(n+1)}{(k+1)!(n+1-(k+1))!} = \frac{(n+1)!}{(k+1)!(n+1-(k+1))!}$

RS $= \binom{n+1}{k+1}$

Pascal's triangle

n=0: 1
n=1: 1 1
n=2: 1 2 1
n=3: 1 3 3 1
n=4: 1 4 6 4 1
n=5: 1 5 10 10 5 1
n=6: 1 6 15 20 15 6 1
n=7: 1 7 21 35 35 21 7 1

rows sums of pascal's triangle

row #		row sum
n=0	1	1
n=1	1 1	2
n=2	1 2 1	4
n=3	1 3 3 1	8

Theorem
for all integers $n \geq 0$.

$$\sum_{i=0}^{n} \binom{n}{i} = \binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{n} = 2^n$$

Proof of theorem (by induction)

For each $n \in \mathbb{N}$, define $P(n)$: "$\sum_{i=0}^{n} \binom{n}{i} = 2^n$"

Base case ($n=0$): LS $= \sum_{i=0}^{0} \binom{0}{i} = \binom{0}{0} = \frac{0!}{0!\,0!} = 1$

RS $= 2^0 = 1$

LS = RS $\therefore P(0)$ holds

Ind. hyp. Assume $P(k)$ holds for some int. $k \geq 0$.
(goal: prove $P(k+1)$ follows)

i.e. assume $\sum_{i=0}^{k} \binom{k}{i} = 2^k$

(goal: prove $\sum_{i=0}^{k+1} \binom{k+1}{i} = 2^{k+1}$)

we have $2^{k+1} = 2 \cdot 2^k$

$= 2\left(\sum_{i=0}^{k} \binom{k}{i}\right)$ by Ind. Hyp.

$= \left(\sum_{i=0}^{k} \binom{k}{i}\right) + \left(\sum_{i=0}^{k} \binom{k}{i}\right)$

$= \left(\binom{k}{0} + \binom{k}{1} + \ldots + \binom{k}{k}\right) + \left(\binom{k}{0} + \binom{k}{1} + \ldots + \binom{k}{k}\right)$

$= \binom{k}{0} + \binom{k}{0} + \binom{k}{1} + \binom{k}{1} + \binom{k}{2} + \ldots + \binom{k}{k-1} + \binom{k}{k-1} + \binom{k}{k} + \binom{k}{k}$

$= \binom{k}{0} + \binom{k+1}{1} + \binom{k+1}{2} + \ldots + \binom{k+1}{k-1} + \binom{k+1}{k} + \binom{k}{k}$

$= 1 + \binom{k+1}{1} + \binom{k+1}{2} + \ldots + \binom{k+1}{k-1} + \binom{k+1}{k} + 1$

$= \binom{k+1}{0} + \binom{k+1}{1} + \ldots + \binom{k+1}{k-1} + \binom{k+1}{k} + \binom{k+1}{k+1}$

$= \sum_{i=0}^{k+1} \binom{k+1}{i}$ $\therefore 2^{k+1} = \sum_{i=0}^{k+1} \binom{k+1}{i}$ so $P(k+1)$ holds!

Ind step complete.

Another proof of theorem 19.2 Let S be an n-element set. $S = \{x_1, \ldots, x_n\}$

Then (# of subsets of S) =

$= \{\underbrace{\emptyset}_{0}, \underbrace{\{x_1\}, \{x_2\}, \ldots, \{x_n\}}_{n}\}$

2 options for x_1 ↑ 2 ways, 2 ways ↑

② $\binom{\text{\# subsets of } S}{} = \binom{\text{\# of subsets containing } x_1}{} + \binom{\text{\# of subsets not containing } x_1}{}$

$= \binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{n}$

① = ② $\therefore \binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{n} = 2^n$

Binomial Theorem $x, y \in \mathbb{R}$, $n \in \mathbb{N}$

$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i$

$= \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \binom{n}{2} x^{n-2} y^2 + \ldots + \binom{n}{n} x^{n-n} y^n$

Binomial coefficients are the coefficients in the expansion of the binomial power

DGD 10

$$(a+b)^n = \sum_{j=0}^{n} \binom{n}{j} a^j b^{n-j} \qquad \left(\frac{3}{x^2} - x^3\right)^8$$

$$\left((-x^3) + \left(\frac{3}{x^2}\right)\right)^8 = \sum_{j=0}^{8} \binom{8}{j} \cdot (-x^3)^j \cdot (3 \cdot x^{-2})^{8-j}$$

$$= \binom{8}{j} \cdot (-1)^j \cdot (x^{3j}) \cdot 3^{8-j} \cdot x^{-16+2j}$$

$$= \binom{8}{j} \cdot (-1)^j \cdot 3^{8-j} \cdot x^{5j-16}$$

i.

x^4

$4 = 5j - 16$ $\qquad \binom{8}{4} \cdot 3^4$

$20 = 5j$

$4 = j$

ii.

x^6

$6 = 5j - 16$

$22 = 5j$

$\frac{22}{5} = j$

iii.

$-1 = 5j - 16$

$15 = 5j$

$3 = j$

$\binom{8}{3} \cdot -1 \cdot 3^5$

$\frac{8!}{3!(8-3)!} \cdot -1 \cdot 3^5$

$\frac{8 \cdot 7 \cdot 6 \cdot 5!}{3! \cdot 5!} \cdot -1 \cdot 3^5$

$8 \cdot 7 \cdot -1 \cdot 3^5$

$8 \cdot 7 \cdot -1 \cdot 3^4 \cdot 3$

$8 \cdot 21 \cdot -1 \cdot 81$

$21 \cdot -81 \cdot 8$

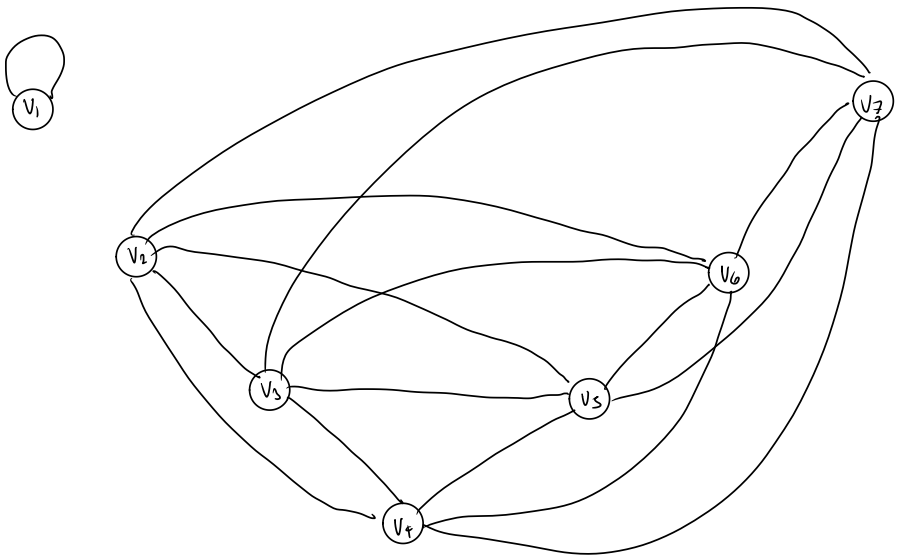
Q2.

a. $(2, 5, 5, 5, 5, 5, 5)$

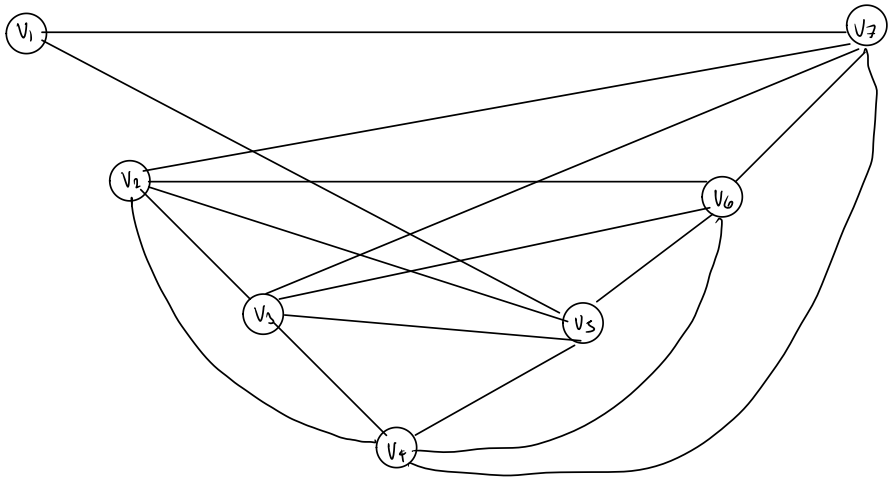
$2 + 30 = 2|E(G)|$

$32 = 2|E(G)|$

$16 = |E(G)|$



graph

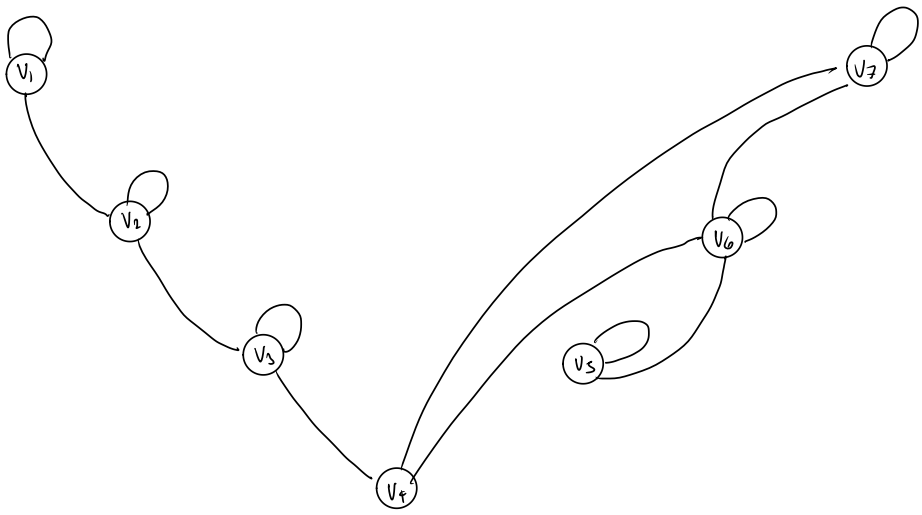


b) $(3, 3, 3, 3, 3, 5, 5)$

$10 + 15 = 2|E(G)|$

$25 = 2|E(G)|$

nope

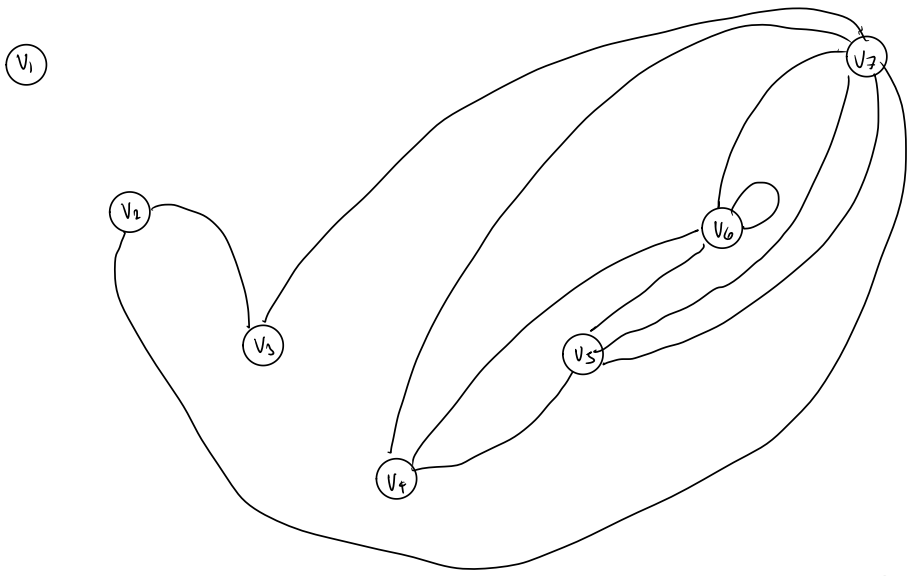


(0, 2, 2, 3, 4, 5, 6)
1 1 1 1 1 1 1

$4 + 8 + 10 = 2|E(G)|$

$11 = |E(G)|$

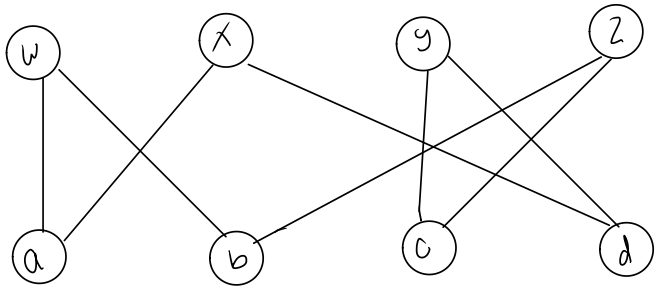
C)



no simple unit graph

Q3.

G1



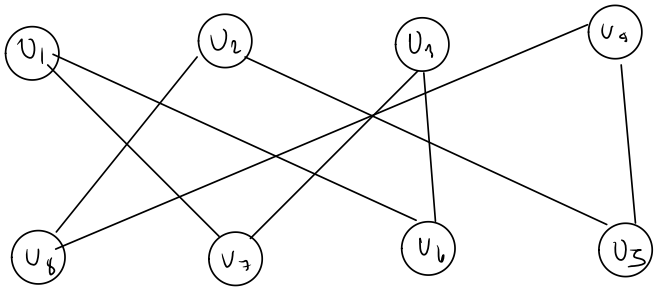
degree sequence: (2,2,2,1,2,2,1,2)

# of vertices: 8

# of edges: 8

adjacency matrix:

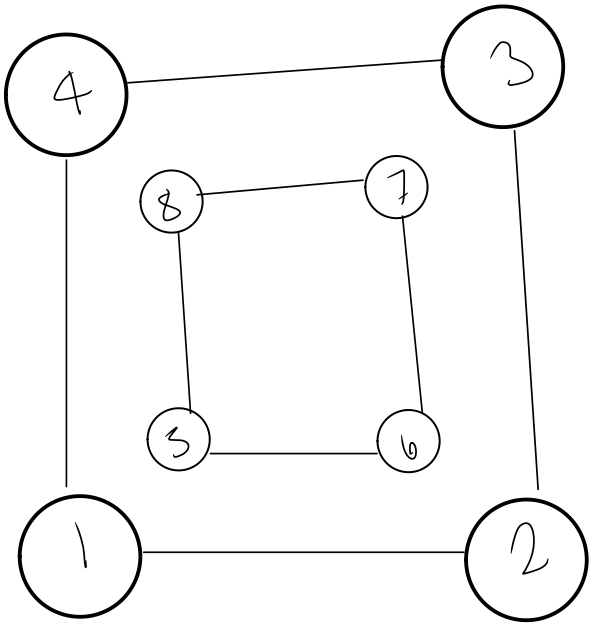
$a_{ij} = 1$



degree sequence: (2,2,2,2,2,2,2,2)

vertices: 8

edges: 8



degree sequence: (2,2,2,2,2,2,2,2)

vertices: 8

edges: 8

$a_0 = 1$

$5^0 \quad 5^1 \quad 5^3 \quad 5^7 \quad 5^{15} \quad 5^{31}$

$P(1) = 5 \cdot (1)^2$

$= 5^1$

$1 + 2 \cdot 0 = 1$

$1 + 2 \cdot 3 = 7$

$1 + 2 \cdot 7 = 15$

11

$P(2) = 5 \cdot 5^2 = 5^3$

$P(3) = 5 \cdot 5^6 = 5^7$

$P(4) = 5 \cdot (5^7)^2 = 5^{15}$

$a_n = 5^{2^n - 1}$

$a_0 = 5^{2^0 - 1} = 1$

$a_1 = 5^{2^1 - 1} = 5$

$a_2 = 5^{2^2 - 1} = 5^3$

$a_3 = 5^{2^3 - 1} = 5^7$

$a_4 = 5^{2^4 - 1} = 5^{15}$

$a_5 = 5^{2^5 - 1} =$

Q1.

5 people

same month 12

same day of week 7

$((4 \cdot 7) \cdot 12) + 1$

$(28 \cdot 12) + 1$

$(14 \cdot 24) + 1$

$(7 \cdot 48) + 1$

$48 \times 7 = 336$

337

Q2.

30 objects Ace → 12 instances

11 or less ↓ 29

$\left\lceil \frac{29}{11} \right\rceil$

$\lceil \quad \rceil$